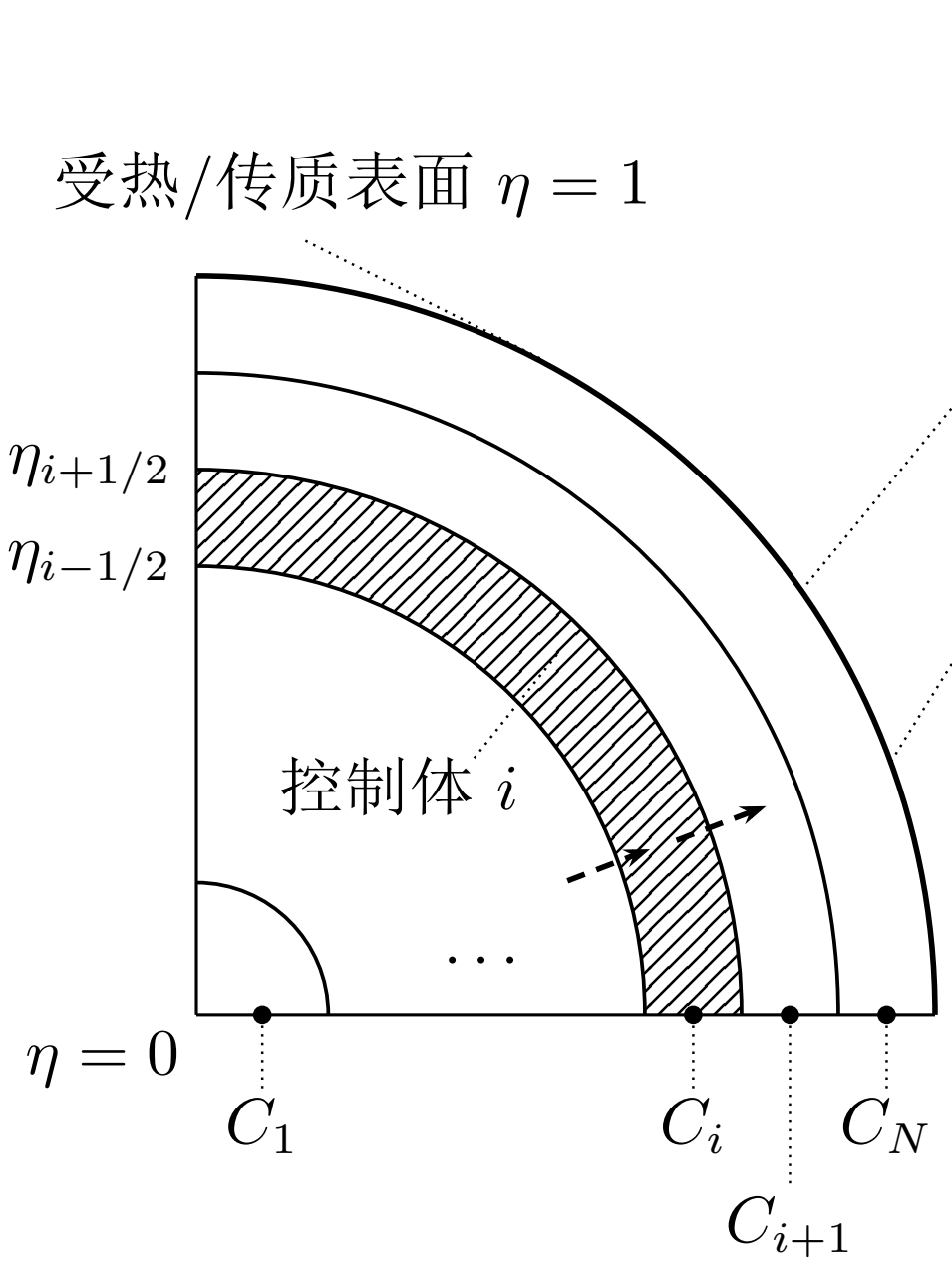
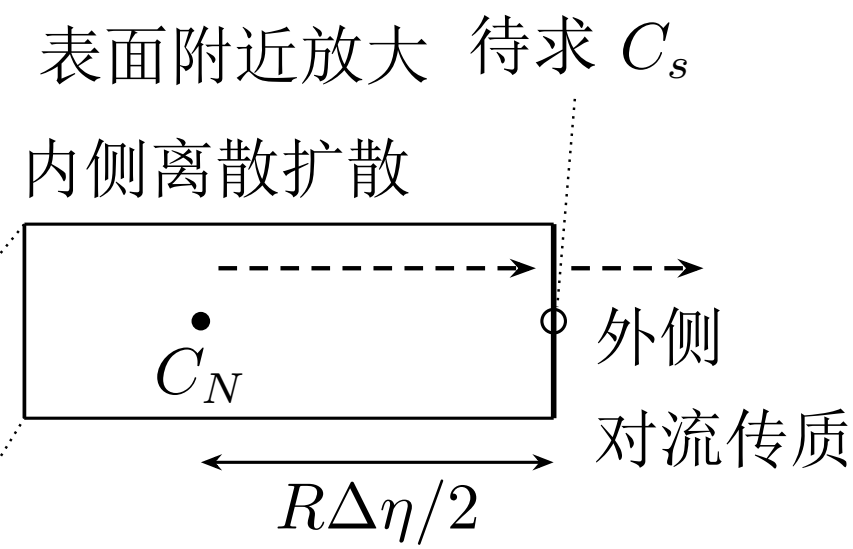


画面环形面积 $\propto$ 体积权

$V_i = (\eta_{i+1/2}^2 - \eta_{i-1/2}^2)/2$

中心面积权 $\eta_{-1/2} = 0$，

轴心零通量自动满足



表面两侧通量相等，消元得

$$C_s = (\beta C_N + h_m C_a)/(\beta + h_m)$$

其中 $\beta = 2D_N/(R\Delta\eta)$

跨界面弧的扩散通量

$$J_{i+1/2} = \eta_{i+1/2} D_{i+1/2} \frac{C_{i+1} - C_i}{\Delta\eta}$$

界面物性取调和平均

控制体 i 的守恒平衡

$$R_0^2 V_i \frac{\mathrm{d}C_i}{\mathrm{d}t} = J_{i+1/2} - J_{i-1/2}$$